CIL viva prep

F1 Race Strategy Intelligence: answers to the 12 questions, using this project's real results

> **HOW TO USE THIS**
> Each question opens with a short answer you can say aloud. The points below it are your elaboration, and **If they push** covers the likely follow-ups. Numbers marked † were measured on 3 Nov 2026 for this prep, with your project's data, splits and pipelines, and are not in `artifacts/`. Unmarked numbers come from `artifacts/`.

The project in 60 seconds

- **Problem:** when should an F1 driver pit, and what will the next lap cost? Two predictions: lap time (regression) and pit at the end of this lap (classification).
- **Data:** the real 2023 Bahrain GP from FastF1. 1,055 laps, 20 drivers, 995 usable rows. Pit laps are 48 of 995, or 4.8%.
- **Split:** laps 4-46 for development (815 rows, 4 expanding-window folds) and laps 47-57 as an untouched test set (180 rows, **only 1 pit lap**).
- **Approach:** a hybrid. Ontology and 32 rules, A* search, classical ML, a neural network, and SHAP/LIME explanations, combined into one recommendation. Decision support for engineers, not an autonomous system.

Numbers to know by heart

	Lap time (test laps)	Pit decision (test laps)
Our ML model	**SVR:** MAE 0.78 s, RMSE 1.03 s, R^2 0.30	**Random forest:** PR-AUC 0.25, ROC-AUC 0.98, F1 0.08, CV PR-AUC 0.39
Our DL model	**MLP [128, 64, 32], 16,257 params:** MAE 0.57 s, RMSE 0.91 s, R^2 0.46	**MLP [16, 8], 289 params:** PR-AUC 0.02, ROC-AUC 0.78, F1 0.00
Better choice	DNN, ahead of SVR by 0.22 s [0.11, 0.33]† per lap	Random forest
Features	45 (28 are driver or team identity)	8 (no identity features)

Three findings that show you understand your results

- **The test set is easy and the "overfitting gap" is mostly one event.** Without laps 40-43, around the VSC period, the DNN's validation MAE is 0.41 s†, below its training MAE of 0.52 s. See Q9.
- **Simple rules are strong competition.** Last lap's field pace gives a test MAE of 0.72 s†, better than SVR on MAE. A tyre-age rule nearly matches the random forest on CV PR-AUC. See Q1.
- **ROC-AUC 0.98 is one lap's rank.** The only test stop is ranked 4th of 180† by the forest, so PR-AUC = 1/4 and ROC-AUC = 176/179. See Q8.

Contents

Q1. Rule-based limits and the case for AI . . . 2
Q2. Adapting to a related application . . . 3
Q3. A changed scenario: is the approach still suitable? . . . 4
Q4. Comparing algorithms . . . 5
Q5. Overfitting and underfitting in the ML models . . . 6
Q6. Feature selection and importance . . . 7
Q7. Distribution shift, imbalance and noise . . . 9
Q8. Evaluation strategy . . . 10
Q9. DL generalisation . . . 11
Q10. Reading the training and validation curves . . . 13
Q11. DL limitations and ways to improve . . . 14
Q12. The CI approach and its suitability . . . 16
Traps and quick definitions . . . 17

Q1. Rule-based limits and the case for AI

Critically examine the limitations of a conventional rule-based approach. Evaluate whether AI provides a meaningful advantage, using the application's characteristics.

> **SAY THIS FIRST**
> Rules are necessary but not sufficient. They handle regulations and rare events well, but they can't predict a continuous lap time from interacting factors. We measured this: the DNN cuts test MAE by 21%† against the best rule. For pit decisions the advantage is much smaller, so we use a hybrid and don't replace rules.

Limitations of rules, from our own system

- **Interactions explode.** Lap time depends on tyre age × compound × track temperature × fuel × traffic. Rules need a threshold for each combination. Our 32 rules cover discrete scenarios and can't output a lap time.
- **Thresholds are brittle.** A tyre-age limit tuned on each fold moved between 11-15 laps (soft) and 13-18 (others)†. Its F1 across folds was 0.55, 0.33, 0.50, then **0.00**†: it caught none of the 6 stops in the last fold.
- **Rules depend on inputs you may not have.** 10 of our 32 rules can't fire in the live pipeline because they need rival gaps, DRS state or a graining model.
- **Coverage is silent.** If no rule fires there is no answer. Safety-car rules fired on 8 laps of the replay, against 31 real SC/VSC laps.
- **Knowledge maintenance.** Every new circuit, regulation or tyre generation means re-authoring rules by hand.

Measured on the same laps

Lap-time approach	CV MAE (s)	Test MAE (s)	Test R^2
Rule: training mean†	2.06	1.76	-1.50
Rule: last lap's field median pace†	1.23	0.72	-0.02
Rule: field pace + fitted tyre-age slope†	1.31	0.79	-0.01
SVR (selected ML model)	1.38	0.78	0.30
DNN	**0.95 (optimistic, see Q4)**	**0.57**	**0.46**

Pit approach	CV PR-AUC	Test PR-AUC	Laps flagged in test
Rule: tyre age, limits tuned on training rows†	0.37	0.05	104 of 180 (F1 0.02)
Random forest (threshold 0.48)	0.39 ± 0.12	0.25	23 of 180 (F1 0.08)

Verdict, by characteristic

- **AI helps** where the target is continuous, the effects are non-linear and data can be refreshed: lap time.
- **AI's edge is unproven** where events are rare: pit timing. The forest is more selective on test, but that rests on 1 pit lap and the CV gap is inside the ±0.12 spread.
- **Rules stay** for regulations, wet-weather changes and safety cars, where there is little or no training data and a decision must be traceable. That is why an expert verdict takes precedence in our recommendation engine.

If they push

"Your SVR loses to a one-line rule on MAE. Why use ML at all?"

Fair challenge. SVR wins on RMSE (1.03 s vs 1.25 s†) and R^2 (0.30 vs -0.02†), so it handles larger swings better. A persistence rule looks good on calm laps because it just lags. The DNN beats it on all three metrics.

"Is 0.15 s per lap meaningful?"

Over a 20-lap stint a systematic 0.15 s is 3 s, which matters against a 22 s pit-loss assumption and undercut margins. But MAE isn't bias, so this is an upper-end illustration and not a measured gain in race time.

Q2. Adapting to a related application

Evaluate adaptability to a different but related scenario. Identify changes in problem characteristics, input information and computational approach, and justify feasibility.

> **SAY THIS FIRST**
> Our chosen extension is endurance racing such as WEC: tyre and fuel stint planning, with many more stops and driver changes. The method transfers, but the trained weights and rule thresholds don't. It's feasible because the pipeline is modular, and because many more pit events per race would ease our biggest data problem. We haven't built it, so this is a reasoned assessment.

What changes

Dimension	Our F1 solution	Endurance racing	Change needed
Problem	Minimise time over fixed laps; 1-3 stops	Maximise distance in fixed time; many more stops; driver swaps; multi-class traffic	New objective and a new goal test in the search
Inputs	Tyre, compound, weather, field pace, gaps; 20 driver and team dummies	Add driver stint, car class, traffic density, fuel and energy limits, day-night swing	New feature blocks; replace name dummies with pace baselines
ML / DL	SVR, RF and MLP on tabular laps	Same start; longer sequences may justify a GRU or LSTM	Retrain; test a sequence model only with enough races
Rules	32 F1 rules	Different regulations and stint limits	Rewrite the rule content; keep the inference engine
Search	A* over lap, compound and stops; 1,005 nodes expanded	Far larger state space	Stronger heuristic, beam search or a rolling horizon

Why it's feasible

- **Reusable as is:** the feature contract, leakage checks, expanding-window validation, threshold tuning, SHAP and LIME tooling, and the API layer. None of these depend on F1 specifics.
- **Low dependence on identity:** driver and team features carry only 3.8% of the DNN's SHAP attribution. Testing on drivers removed from training raised SVR error modestly, from 0.77 s to 0.86 s†, using race-state features only.
- **Better class balance:** more stops per car means more positive examples, the main weakness of our pit model.
- **Main risk:** data access. FastF1 serves F1 only, so a new timing source and data contract are required.

If they push

"What's the easiest extension?"

Other F1 circuits. FastF1 uses the same schema, so it needs a fetch and retrain, plus circuit features. It's also the best next step for this project, since it fixes the single-race limit.

"Would transfer learning work?"

Plausibly: pretrain the MLP on F1, then fine-tune the upper layers. We haven't tested it, and with so little data a different lap-time scale and tyre behaviour could cancel the benefit.

"What wouldn't transfer at all?"

The trained weights, thresholds such as 0.48, the fuel coefficient of -0.0613 s per lap, and the rule content.

Q3. A changed scenario: is the approach still suitable?

Consider a change in the problem scenario, input conditions or application environment and determine whether the existing approach remains suitable.

> **SAY THIS FIRST**
> The hybrid architecture remains suitable. The trained models do not, once conditions leave what one dry evening race showed them. In that case ML should be flagged or disabled, rules and search carry the decision, and the models need retraining. A race 5 °C cooler puts every test lap's temperature feature outside its training range†, and DNN error rises from 0.57 s to 4.5 s†.

Changes tested or assessed

Change	What happens	Evidence	Still suitable?
Track 5 °C cooler (trained on 28.7-31.2 °C only)	Temperature-tyre feature is outside its training range on 100% of test laps†	MAE: SVR 2.02 s, DNN 4.50 s†. The full pipeline's call flips on 7 of 9 trials	No, retrain
Track 5 °C hotter	Same out-of-range problem	SVR 1.96 s; DNN 0.52 s†. Don't read that as robust, the model is extrapolating	No
Rain	`is_rainfall` was removed for zero variance, and there are no wet compounds	0 of 1,055 laps had rain, so ML can't see it at all	ML no; rules yes (wet-changeover scenario)
Safety car or VSC	A regime almost absent from training	DNN MAE 2.77 s on non-clear laps, 1.06 s on clear laps†	Partly; rules take precedence
New circuit	Pace baseline changes; lag features adapt only partly	Not tested; 1 circuit available	Retrain
Live deployment	Some inputs are approximated; sensor noise	p50 latency 101 ms, p95 125 ms, against a ~98 s lap	Yes on speed; see Q7 on noise

What already protects us

- **Out-of-range detection:** any derived feature outside the training min and max raises an "extrapolating" warning in the response and dashboard.
- **Rule precedence:** an expert-system verdict overrides ML and DL.
- **Human oversight:** the report states this is decision support for engineers, not autonomous.

What we would add

- **Drift monitoring:** KS tests on inputs plus a rolling MAE, with a retraining trigger.
- **An explicit fallback mode:** when too many features are out of range, return the rule and search answer and label ML as unavailable.
- **A training set that spans conditions:** several races, including wet and high-temperature ones.

If they push

"Doesn't your trust score handle this?"

No, and we say so. Its correlation with lap-time error is -0.09 ($p = 0.24$). It's a project-defined summary of agreement and input validity, so read its components.

"Why did DNN error fall at +5 °C?"

Probably coincidence. Temperature fell through the race, so test laps sit at the low end of what the model saw, and shifting up moves them back toward familiar values. It isn't evidence the DNN handles heat.

Q4. Comparing algorithms

Analyse the effect of different ML algorithms. Compare your model with an alternative on predictive performance, interpretability, training requirements and computational complexity, and justify the more appropriate choice.

> **SAY THIS FIRST**
>
> It depends on the target. For lap time the DNN is the better choice: MAE 0.57 s against SVR's 0.78 s, a difference that stays positive when we resample whole laps. For pit decisions the random forest wins clearly, because 41 positive training laps aren't enough for a network. We use the forest there, and we report that the DNN lost.

Criterion	SVR (lap time)	DNN (lap time)	Random forest (pit)	DNN (pit)
Test performance	MAE 0.78, R^2 0.30	MAE 0.57, R^2 0.46	PR-AUC 0.25; ranks the stop 4th†	PR-AUC 0.02; ranks the stop 41st†
CV score	MAE 1.38 ± 0.56	MAE 0.95 (optimistic)	PR-AUC 0.39 ± 0.12	PR-AUC 0.48 ± 0.12 (optimistic)
Interpretability	Low; RBF kernel has no native importances	Low; needs sampled SHAP	Medium; importances and SHAP	Low
Size	724 support vectors†	16,257 params; 167 KB file	300 trees, 10,830 nodes†	289 params
Fit time	0.03 s†	About 11 s per fit†	0.23 s†	Not measured
Tuning effort	Grid of 8 configs	13 configs, 4 folds each	Grid of 6 configs	13 configs, 4 folds each
Training complexity	Between $O(n^2)$ and $O(n^3)$	O(epochs × n × params)	O(trees × m × n log n)	Same as DNN
Prediction cost	O(support vectors × d); 5.6 ms per 180 laps†	O(params)	O(trees × depth); 27 ms per 180 laps†	O(params)

Why the winner differs by target

- **Lap time:** every row gives a continuous error signal, so 675 rows can train 16,257 parameters with dropout, L2 and early stopping. The paired difference (SVR minus DNN) is 0.22 s, with a 95% lap-block bootstrap interval of [0.11, 0.33]†.
- **Pit decision:** only 41 training rows are positive. Trees isolate tyre-age regions from few examples, while the 289-parameter network still reaches only ROC-AUC 0.68 on its own training data, so it underfits.
- **Quantum models:** on the same split they reached CV PR-AUC 0.30 and lap-time CV MAE 1.75 s. That is a simulated 4-qubit comparison and shows no quantum advantage.

If they push

"The DNN's CV PR-AUC of 0.48 beats the forest's 0.39. Why choose the forest?"

Those CV figures aren't like for like. The network's early stopping watches the same validation fold it's scored on, which flatters it. Scikit-learn models never see that fold. On the shared test laps the forest ranks the stop far higher, though that's only one lap.

"Why was SVR selected over the decision tree, which has better CV MAE (1.19 s)?"

The tree's test R^2 is -0.17, worse than predicting the mean. Our generalisation guard refuses a negative-R^2 model when a positive one exists. It's a judgement, and it does use test information, so we disclose it in the selection report.

"Does your deployed system follow this?"

Mostly. The pipeline compares both families and falls back to ML on pit disagreement. Be aware that the ML pit class currently uses scikit-learn's 0.5 cut-off, not the tuned 0.48 (`strategy_service.py:172`). It's a small known gap.

Q5. Overfitting and underfitting in the ML models

Analyse possible overfitting or underfitting using training and testing results. Evaluate causes and recommend corrective strategies, justifying their expected impact.

> **SAY THIS FIRST**
> Our flexible models overfit, and the train-to-test gap shows it: SVR falls from R^2 0.89 to 0.30, and the random forest's PR-AUC falls from 0.80 to 0.25. Neither underfits. The causes are few rows, many sparse features and a shift between early and late laps. The most useful fixes are removing identity features, reducing SVR's C and, above all, adding more races.

Train, CV and test†

Lap-time model	Train MAE	Train R^2	CV MAE	CV R^2	Test MAE	Test R^2	Reading
Linear regression	0.98	0.47	2.02	-1.28	0.79	0.14	Limited and unstable
Decision tree	0.39	0.89	1.19	0.34	0.87	-0.17	Overfit
Random forest	0.24	0.96	1.38	0.21	0.93	-0.06	Strongly overfit
XGBoost	0.37	0.92	1.44	0.14	1.09	-0.37	Strongly overfit
SVR (selected)	**0.29**	**0.89**	**1.38**	**0.26**	**0.78**	**0.30**	**Overfit, but generalises best**

Pit model	Train PR-AUC	CV PR-AUC	Test PR-AUC	Reading
Logistic regression	0.39	0.30	0.25	Smallest gap; little overfitting
Random forest, depth 5 (selected)	0.80	0.39	0.25	Overfit, but best CV
Random forest, unlimited depth	1.00	0.32	0.20	Memorises training data
XGBoost	0.98	0.31	0.07	Strongly overfit

Causes

- **Few rows for the features:** 815 rows and 45 features, of which 28 are identity dummies. The first CV fold trains on only 180 rows, which is why its error is high.
- **High-capacity settings:** with C = 10 and ε = 0.05, 724 of 815 rows (89%) become support vectors†. The model is close to storing its data.
- **Time shift:** early laps are heavy-fuel and hot; late laps are light and cool. Track temperature × tyre life has a KS distance of 0.92 between development and test laps†.
- **Rare positives:** 48 pit laps in total.

Corrective strategies and measured effect†

Strategy	Measured effect	Justification
Remove identity features (SVR, 45 → 17)	CV MAE 1.38 → 1.28; test MAE 0.78 → 0.73; test R^2 0.30 → 0.32	Better on all three, with far fewer inputs. Our top recommendation
Reduce SVR C from 10 to 1	Train R^2 0.89 → 0.67; test MAE 0.78 → 0.62; test R^2 ≈ same (0.30); CV MAE 1.38 → 1.39	Narrower gap at equal CV. Prefer the simpler model when CV can't separate them
Limit forest depth	Unlimited: CV 0.32. Depth 5: CV 0.39. Depth 3, leaf 10: CV 0.29	Depth 5 is about the sweet spot; tightening further starts to underfit
More races	Not measured; expected to help most	Addresses the root cause: rows, positives and conditions
Keep the expanding-window CV and the selection guard	Already in place	Stops us picking a model that only looks good on one stretch of laps

If they push

"SVR's test MAE (0.78) is better than its CV MAE (1.38). Doesn't that disprove overfitting?"

No. The test laps are simply easier: lap-time SD is 1.23 s, against 2.74 s in the last validation block†, and they contain no VSC laps. Folds 1 and 4 (1.87 s and 2.01 s) drive the CV mean, while folds 2 and 3 are 0.77 s and 0.88 s. R^2, which accounts for that spread, falls from 0.89 to 0.30.

"Is anything underfitting?"

Linear regression has limited capacity (train R^2 0.47) and is unstable across folds. The pit DNN also underfits; see Q10.

Q6. Feature selection and importance

Evaluate the influence of feature selection and importance. Identify irrelevant or redundant features and justify how removal or prioritisation could influence effectiveness and interpretability.

> **SAY THIS FIRST**
>
> A small group of race-state features does most of the work, led by tyre life, the temperature-tyre interaction, recent form and field pace. Most of the 45 features add little. 32 of them never reached a top-12 list in any fold, and removing the 28 identity features actually improved SVR on CV and test†. Fewer features also make explanations more honest and easier to act on.

How selection worked

- **Funnel:** 50 candidates, then near-zero variance (removed `is_rainfall`), correlation above 0.95 (`tyre_life_sq`, `air_temp`, `track_air_delta`), VIF above 10 (`race_progress`, VIF 24.9), then importance and stability.
- **Ranking:** average rank of mutual information, random-forest importance and L1 coefficient. **Stability** is the share of folds in which a feature reached the top 12.
- **Result:** 45 features for lap time, 8 for pit decisions. Sector times and speed traps were excluded earlier as leakage.

What matters

Target	Most important (permutation, DNN)	Least useful
Lap time	`tyre_life` 0.160, `tracktemp_dev_x_tyrelife` 0.140, `form_vs_baseline` 0.111, `gap_roll3_mean` 0.110	Driver dummies, `compound_medium`, `tyrelife_x_medium`, `humidity`
Pit decision	`tyre_life` 0.245, `tyrelife_x_soft` 0.190, `tracktemp_dev_x_tyrelife` 0.162	`gap_roll3_mean` -0.011, `field_pace_trend` -0.002

Irrelevant or redundant, and why

- **Identity dummies (28):** 27 have stability 0, and together they hold 3.8% of SHAP attribution though they are 62% of features. They also can't generalise to a new driver.
- **Medium-compound features:** only 10 of 1,055 laps used mediums, and none are in the test laps.
- `humidity`**:** nearly constant (SD 0.34). `track_status`**:** constant in most training windows and mis-scaled (Q9).
- **Redundant pairs:** `compound_medium` and `tyrelife_x_medium` (r 0.89), `compound_soft` and `tyrelife_x_soft` (r 0.85), `stint_number` and the temperature interaction (r 0.79)†.

Effect of removal, using the project's pipeline†

Feature set	k	SVR CV MAE	SVR test MAE	SVR test R^2	RF CV MAE	RF test MAE
All features	45	1.38	0.78	0.30	1.38	0.93
No driver dummies	26	1.31	0.82	0.24	1.42	0.96
No identity (race state only)	17	**1.28**	**0.73**	**0.32**	1.40	0.98
Top 12	12	1.30	0.88	0.16	1.43	1.06
Top 6	6	1.23	1.09	-0.24	2.24	0.67
Without `tyre_life`	44	1.43	0.79	0.29	2.22	0.60

- **Removal helps up to a point.** Going to 17 helps; at 6, SVR's CV improves but test collapses. Don't select on a single metric.
- **DNN agrees:** without identity features, CV MAE 1.06 vs 1.05 and test MAE 0.57 vs 0.59†, within seed noise, and with 3,584 fewer parameters.
- **Pit model:** dropping the two lowest features lowers CV PR-AUC from 0.39 to 0.31†, even though their permutation scores are about zero. Those scores rest on 1 positive lap, so we keep the features.

If they push

"`tyre_life` has an L1 coefficient of 0. How can it be the most important?"

L1 shares credit among correlated features: its interactions and stint number absorb its linear signal. Tree importance (0.275) and permutation rank it first. That's why we average several criteria.

"Why keep 45 if 17 is better?"

The Task 5 rule keeps the smallest K within 2% of the best CV MAE under its own model: 45 scored 0.99, and 20 scored 1.03. That rule retained them, and the README flags the risk. With hindsight we'd select by stability and drop identity features.

"LIME ranks `track_status` first. Is it important?"

No. It's constant on all test laps, so that's an explainer artefact. We report LIME as unreliable here and rely on SHAP and permutation importance.

Q7. Distribution shift, imbalance and noise

Analyse how changes in data distribution, class imbalance or noisy data could affect performance. Recommend strategies and justify them with reference to the problem.

> **SAY THIS FIRST**
> All three are real in this project. Distribution shifts within a single race as fuel burns and the track cools. Pit laps are only 4.8% of the data. Timing data carries capped outliers and label errors. We measured each†: shift does the most damage, feature noise hurts steadily, and the models tolerate missing pit labels fairly well.

Distribution shift

- **Inside one race:** KS distance between development and test laps is 0.92 for temperature × tyre life, 0.76 for field pace, 0.61 for stint number and 0.53 for tyre life†. Mean tyre life goes from 8.7 to 17.1 laps†.
- **Effect:** CV rankings and test rankings disagree. Decision tree is best on CV and has negative R^2 on test.
- **Regime change:** VSC laps have a DNN MAE of 2.77 s, against 1.06 s on clear laps†.
- **Strategies:** expanding-window validation, a multi-race training set, drift monitoring, and regular retraining. Race-state features such as field pace also adapt better than fixed names do.

Class imbalance (random forest, same folds†)

Method	CV PR-AUC (mean ± SD)	Comment
No weighting	0.34 ± 0.13	Baseline
Class weight "balanced" (ours)	**0.39 ± 0.12**	Best here, simple, no synthetic rows
Random oversampling to 30%	0.34 ± 0.11	No gain
Undersampling to 1:5	0.30 ± 0.09	Throws away data

- **These gaps are inside the spread.** The folds hold only 6-11 positives each, so the ranking is suggestive only.
- **Accuracy is meaningless here:** always predicting "no pit" scores 95.2%.
- **Our recommendation:** class weighting, a threshold tuned on out-of-fold predictions (0.48 rather than 0.5), PR-AUC for selection, and more races for real positives.
- **DNN note:** `fit(class_weight=...)` was found to have no effect on this Keras/PyTorch install, so weights go inside the loss. Weighting then lowered CV PR-AUC (0.36 vs 0.42), so the final network doesn't use it.

Noise†

Gaussian noise on test features (× feature SD)	0	0.1	0.25	0.5	1.0
SVR test MAE (s)	0.78	0.80	0.90	1.11	1.50
DNN test MAE (s)	0.57	0.58	0.68	0.92	1.56
Random forest test ROC-AUC	0.98	0.92	0.78	0.74	0.78

Noise in training labels	None	Light	Medium	Heavy
Lap time ± SD of 0, 0.5, 1, 2 s: SVR test MAE	0.78	0.81	0.89	1.24
0, 10, 25, 50% of pit labels missed: RF CV PR-AUC	0.39	0.35	0.35	0.32

- **Real noise in our data:** 79 laps capped at 103.985 s by the IQR rule (60 are in- or out-laps), NOR's stop on lap 47 is unlabelled because stint data was median-imputed, and FastF1 tyre age sometimes exceeds the lap number.
- **Strategies:** fix NOR's label at the source, flag in- and out-laps in place of capping them, use robust rolling features, and keep input validation. Huber loss gave no clear benefit (CV MAE 1.06 vs 1.05†).

- **DNN vs SVR:** the DNN stays ahead until very heavy noise, when they converge.

If they push

"Why not SMOTE?"

Pit stops depend on strategy and timing, so an interpolated row may describe a race state that never happened. Simple oversampling already gave no gain, and SMOTE here would need real evidence. We didn't test it.

"Doesn't capping hide real behaviour?"

Yes. The target can't exceed 103.985 s, which flatters error on the slowest laps. A better design treats in- and out-laps as their own regime.

Q8. Evaluation strategy

Determine an appropriate evaluation strategy to assess reliability under the considered data conditions. Justify validation techniques and metrics with respect to the problem.

> **SAY THIS FIRST**
> We use expanding-window lap-forward cross-validation plus an untouched chronological holdout, because this is time-ordered data and random K-fold would train on the future. We select with MAE for lap time and PR-AUC for pit decisions, because pit laps are rare. Since the holdout has only one stop, we also rely on pooled out-of-fold results and bootstrap intervals.

Validation design

Fold	Train laps	Validate laps	Train rows	Val rows	Pit laps in train / val†
1	4-12	13-21	180	172	11 / 10
2	4-21	22-30	352	171	21 / 11
3	4-30	31-38	523	152	32 / 9
4	4-38	39-46	675	140	41 / 6
Final test	Laps 4-46 for selected ML model	47-57	815	180	48 total / 1 in test

- **No random K-fold:** it would train on lap 50 and validate on lap 10. Lag features would also leak.
- **Whole laps stay together**, so the same race moment can't appear on both sides.
- **Scalers are fit inside each fold**, and thresholds are tuned on out-of-fold predictions.
- **Leakage checks:** a test fails the build if sector times, speed traps or `IsPersonalBest` reach a training matrix.

Metrics and why

Metric	Why it fits	Watch out
MAE (primary, lap time)	In seconds; a direct strategy cost; robust to outliers	Flattered by calm laps
RMSE and R^2	Penalise large misses; R^2 compares with predicting the mean	R^2 depends on the variance of that block
PR-AUC (primary, pit)	Focuses on the rare class; baseline equals the positive rate (about 5.7% in CV)	Very noisy with few positives
ROC-AUC	Ranking quality, reported for reference	Flattering under imbalance
Precision, recall, F1	The operating point after threshold tuning	With 1 positive, recall can only be 0 or 1
Accuracy	Not used for selection	95.2% for a useless model

Why the test pit numbers can't be taken at face value

- With a single positive, **PR-AUC = 1 / rank**. The forest ranks the stop (lap 54, tyre life 25) 4th†: 1/4 = 0.25. The DNN ranks it 41st†: 1/41 = 0.024.
- **ROC-AUC is the same information:** (179 - 3) / 179 = 0.983, and (179 - 40) / 179 = 0.777.
- **A better estimate:** pooled out-of-fold predictions have 36 positives in 635 rows. PR-AUC 0.19, 95% bootstrap interval [0.11, 0.31]†, against a base rate of 0.057. Precision 0.25, recall 0.36†, with the threshold tuned on those same rows, so slightly optimistic.

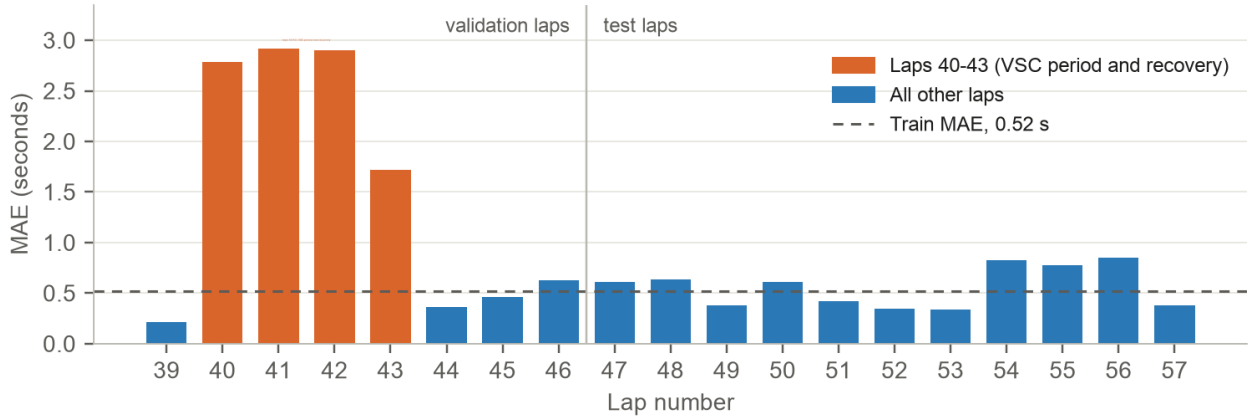
How we check reliability beyond a single number

- **Baselines:** every model is compared with naive rules on the same laps (Q1).
- **Uncertainty:** lap-block bootstrap, because laps within a driver are correlated, and repeated seeds. The DNN's test MAE is 0.59 ± 0.04 over 3 seeds†.
- **Stratification:** error by team, driver, compound and regime. McLaren has 2.32 s, against 0.27 s for Red Bull, but that's 9 laps and includes NOR's label gap.
- **Robustness:** perturbing inputs flips the pipeline's call on 7 of 9 trials at -5 °C and at tyre age -2, and on none at +5 °C or +2.
- **Domain replay:** replaying the race gives recall 0.87 (34 of 39 stops) and precision 0.125 (237 false calls). Most replayed laps were in the training range, so treat this as a plausibility check, not generalisation evidence.
- **Next step:** leave-one-race-out validation, the correct test once there are more races.

Q9. DL generalisation

Analyse the generalisation capability of your DL model using training and validation performance. Determine whether it overfits or underfits, and propose a suitable modification technique.

> **SAY THIS FIRST**
>
> The lap-time network shows moderate overfitting, and regularisation is controlling it. But its large train-to-validation gap (0.52 s vs 1.50 s) is mostly a regime problem: excluding laps 40-43, validation MAE is 0.41 s†. Training laps never contained a VSC. So our main modification is to fix how track status is encoded and refit on all development laps. That improved test MAE from 0.59 s to 0.51 s†.



Saved DNN, absolute error per lap†. The validation gap comes from four laps; the rest of validation looks like the test laps.

Reading the evidence

Split (laps)	MAE (s)	RMSE (s)	R^2	Target SD (s)†
Train (4-38)	0.52	0.86	0.78	1.85
Validation (39-46)	1.50	2.38	0.25	2.74
Validation without laps 40-43†	0.41			
Test (47-57)	0.57	0.91	0.46	1.23

- **Real overfitting exists:** after epoch 10, training loss falls from 0.47 to 0.27 while validation loss rises 40.6%, and R^2 drops from 0.78 to 0.46 on test.
- **It isn't underfitting:** train R^2 is 0.78, and a smaller network does worse (CV MAE 1.26 vs 1.05†).
- **Regularisation works:** without dropout and L2, test MAE is 1.22 s and validation MAE 2.73 s†, against 0.59 s and 1.53 s with them.

The encoding defect behind the curves

- `track_status` holds FastF1 status strings as numbers: 1, 12, 126, 671, 2671. They're codes, not magnitudes. Status is non-clear on laps 39-42†, and the error spike follows on laps 40-43.
- In the final fit's 675 training rows it is always 1, so its SD is 0 and the scaler divides by 1†. Validation rows then enter the network with values up to **2,670**†.
- That is why the first epoch's validation loss is 71, or 1,290 with another seed†, against about 2-4 once the column is fixed†. Its weights get no signal from the training data, so L2 shrinking them is most likely what brings validation loss down.

Proposed modification and measured effect†

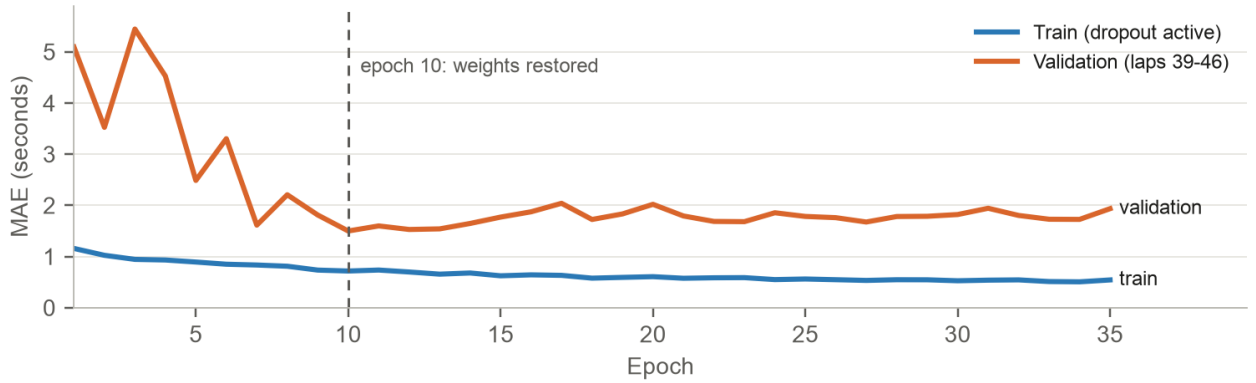
Variant (mean of 3 seeds)	Best epochs	Test MAE (s)	3-seed ensemble
Selected network, as built	10, 36, 6	0.59 ± 0.04	0.53
Remove `track_status`, same early stopping	5, 1, 5	0.73 ± 0.04	
Binary non-clear flag, same early stopping	5, 1, 6	0.71 ± 0.07	
Binary flag, stop on clear laps only	56, 42, 68	0.60 ± 0.06	
Binary flag, refit on all 815 rows, 20 epochs	fixed	**0.51 ± 0.04**	**0.49 (R^2 0.52)**

- **The fix alone isn't enough.** With a clean curve, early stopping fires at epochs 1-6 because the validation block is dominated by a regime the network never saw. So it stops too early.
- **Refitting on all development laps** lets the network actually see VSC laps and uses 21% more data.
- **Honest caveat:** we picked 20 epochs after seeing test results (40 epochs gave 0.56 ± 0.08), so this is illustrative. The proper version chooses the epoch count from CV, then re-tests on a new race.

Q10. Reading the training and validation curves

Analyse the training and validation performance of your existing model to identify possible overfitting or underfitting.

> **SAY THIS FIRST**
> The lap-time network overfits after epoch 10, and early stopping restores that epoch. The pit network has the opposite weakness: the project's rule labels it a reasonable fit, but its train ROC-AUC of 0.68 and PR-AUC of 0.18 show it hasn't learned the task well, so it leans towards underfitting. For the classical models, use the table in Q5.



Lap-time DNN, in seconds (the history is in standardised units; multiply by the training SD of 1.85 s).

Lap-time network, epoch by epoch

Epoch	Train loss	Val loss	Train MAE (s)	Val MAE (s)	What's happening
1	0.98	71.46	1.16	5.10	Mis-scaled `track_status` dominates validation
6	0.63	18.51	0.85	3.30	Validation still falling fast
10	**0.47**	**1.79**	**0.72**	**1.50**	**Best epoch; weights restored**
16	0.38	2.63	0.64	1.87	Train improves, validation worsens: overfitting starts
35	0.27	2.52	0.54	1.94	Patience of 25 runs out

- **Our rules, stated in `dl/pipeline.py`:** overfitting emerged if validation loss rose more than 10% above its minimum while training loss kept falling. The saved model is overfit if train MAE is below 60% of validation MAE. Underfit if train R^2 is below 0.1.
- **Result:** +40.6% rise, and 0.52 is 35% of 1.50, so the verdict is overfitting.
- **Why loss and MAE look different:** training metrics during epochs include dropout, so they read higher than the 0.52 s at saved weights. Loss is in standardised units and includes the L2 penalty.

Pit network

Split	Pit laps	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Train	41 / 675	0.84	0.15	0.37	0.21	0.68	0.18
Validation	6 / 140	0.96	1.00	0.17	0.29	0.97	0.61
Test	1 / 180	0.99	0.00	0.00	0.00	0.78	0.02

- **Underfitting signs:** weak on its own training rows, with a recall of 0.37. Validation scoring above training isn't skill: it's 6 positives, and dropout is off.
- **Overfitting after epoch 10 too:** validation loss rises 29.9% from 0.126 to 0.164.
- **Why the loss looks fine:** with few positives a low loss comes from predicting "no pit". Validation accuracy settles at 0.957, which is 134/140, the majority class.

- **Conclusion:** it's a data problem more than a tuning one. More capacity wouldn't fix 41 positives (with other settings equal, [128, 64, 32] scored CV PR-AUC 0.42 against 0.45 for [16, 8]).

If they push

"Why does validation loss start at 71 when training loss starts at 1?"

A mis-scaled `track_status` input, not overfitting. See Q9.

"Why is the pit train loss higher than validation at epoch 10 (0.25 vs 0.13)?"

Dropout and L2 are active on the training loss, and the validation block has fewer positives (4.3% vs 6.1%).

"Why stop at epoch 10 and not continue?"

Validation loss didn't improve for 25 epochs, so patience ran out at epoch 35, and `restore_best_weights` brought back epoch 10.

Q11. DL limitations and ways to improve

Identify possible limitations in generalisation capability. Determine suitable modification techniques to improve its ability to generalise to unseen data.

> **SAY THIS FIRST**
>
> The biggest limit is that the network has only seen one race, so nothing we do inside this dataset can make it generalise to other circuits. Within the data, the most effective changes we measured are refitting on all laps with a fixed status flag, and averaging several seeds. Most standard tweaks, like a smaller network or more dropout, did not help, and we can show that.

Limitations and suitable modifications

Limitation	Evidence	Modification	Status
One race, one circuit	28.7-31.2 °C; dry; 180 test laps	Train on many races; validate leave-one-race-out	Highest expected gain; not measured
Regime unseen in training	VSC laps 2.77 s vs clear 1.06 s†	Fixed flag and refit on all rows	0.59 → 0.51 s†
Sensitive to random seed	Best epochs 10, 36, 6; test SD 0.04 s†	Average 3-5 seeds	0.59 → 0.53 s†
Driver and team names	28 features with 3.8% of attribution	Drop them, or use pace baselines or embeddings	CV unchanged; test 0.59 → 0.57 s†
Extrapolation behaves arbitrarily	-5 °C: 4.5 s†; older tyres predicted faster on out-laps	Physics-informed loss or monotonic constraints; out-of-range gate	Gate built; constraints not tested
No memory of the stint	Plain MLP; only lag features	GRU or LSTM per driver stint	Not tested; needs far more data
Capped target	79 laps at 103.985 s	Separate in- and out-lap regime	Not tested
Few pit positives	41 in training; test PR-AUC 0.02	More races, or keep the forest for this task	Forest in use

What didn't help, which is worth saying†

Variant (3 seeds, 4 folds)	CV MAE (s)	Train MAE	Test MAE	Reading
Selected (dropout 0.2, L2 0.001)	1.05 ± 0.07	0.53	0.59	Baseline
No dropout or L2	1.41 ± 0.33	0.41	1.22	Clearly worse: regularisation is earning its place
Higher L2 (0.01)	1.03 ± 0.02	0.41	0.63	Within noise; more stable across seeds
Higher dropout (0.4)	1.06 ± 0.05	0.56	0.74	No gain
Smaller network [32, 16]	1.26 ± 0.06	0.66	0.80	Worse: capacity isn't the problem
Huber loss	1.06 ± 0.07	0.40	0.59	No gain
No identity features	1.06 ± 0.06	0.48	0.57	Same accuracy with fewer inputs

Note: the 3-seed CV of 1.05 s is the fairer number. The report's 0.95 s came from a single seed during tuning, and CV for the DNN is optimistic in any case, because early stopping uses the validation fold (Q4).

Priority order to give if asked

- **1. More races**, with leave-one-race-out validation. It addresses the root cause.
- **2. Fix** `track_status` and make sure training windows include non-clear laps.
- **3. Seed ensemble** for a cheap, reliable gain.
- **4. Drop identity features** for simplicity and fairness.
- **5. Add physics constraints**, so tyre age can't make a car faster, before considering sequence models.

If they push

"Why not batch normalisation or data augmentation?"

Neither addresses the actual problem. Batch norm is helpful for deep networks, but ours is shallow and small. Augmenting tabular race data risks inventing states; adding real races is safer.

"Why not use an LSTM straight away?"

It has more parameters, and we only have about 50 laps per driver. Lag and rolling features already provide a short memory. Compare it to the MLP once there are many races.

Q12. The CI approach and its suitability

Identify the computational intelligence approach adopted and justify its suitability, considering problem characteristics, application requirements and expected performance.

SAY THIS FIRST

We use a hybrid approach. A neural network and classical learners provide the data-driven prediction, a rule-based expert system and ontology provide domain knowledge, and A* search plans the strategy. No single technique fits, because this problem mixes noisy continuous prediction, rare events, hard rules and sequential planning. Engineers also need to see why a call was made.

Problem characteristic to technique

Characteristic	Technique	Why it suits	Performance
Non-linear, noisy, continuous lap time	Neural network (MLP)	Learns interactions with no hand-written equation	MAE 0.57 s, R^2 0.46
Rare events; small data; tabular	Random forest	Works with few positives; class weighting	CV PR-AUC 0.39, about 7× the 0.057 base rate
Regulations and expert knowledge; rare situations	Ontology (61 entities, 29 relations) and 32 rules with forward and backward chaining	No training data needed; each firing has an explanation	6/6 validation checks; 22 of 32 reachable live
Sequential strategy with a cost	A* search	Provably optimal with an admissible heuristic	2,262.42 s, same as UCS, 1,005 vs 1,589 nodes
Trust and accountability	SHAP, LIME, counterfactuals	Show why; exposed non-physical behaviour	Top SHAP feature stable in 92-100% of resamples
Exploration	Quantum ML (simulated, 4 qubits)	A fair, documented comparison	No advantage; reported as such

Requirements it has to meet

- **Low latency:** p50 101 ms, p95 125 ms without explanations; 258 ms and 292 ms with them. Laps take about 98 s, so this leaves plenty of margin.
- **Traceability:** recommendations carry their reason, source, confidence, fired rules, and ML and DL evidence. The automated traceability check passed.
- **Graceful behaviour:** if a model is missing, the response says "not available" and never uses a made-up value. Disagreement between models is reported, not hidden.
- **Combining rule:** an expert verdict wins if one fires. Otherwise ML and DL are compared; if they disagree, ML breaks the tie and confidence is marked low.

How it fits the classic definition of CI

- CI classically means **neural networks, fuzzy systems and evolutionary computation**. Our CI core is the neural network, and the rest is wider AI: symbolic reasoning and search.
- **We don't use fuzzy logic or evolutionary algorithms.** Rules use crisp conditions with confidence values. Fuzzy membership for "high wear" is a natural extension, and a genetic algorithm would make sense if the strategy space outgrew A*.
- **Why A* now:** the state space is small enough for a guaranteed optimum (about 14 ms), so a GA's approximate answer would give no benefit.

Expected performance, honestly stated

- **Good:** sub-second average lap-time error on normal laps of this race, with transparent reasoning.
- **Weak:** pit call precision (0.125 in the replay), so it's a prompt for an engineer, not an instruction. Robustness to input changes is also limited (Q8).
- **Unknown:** any other race. All results come from one Grand Prix.

If they push

"Why not just use the neural network for everything?"

It failed on pit decisions (PR-AUC 0.02), can't encode regulations and doesn't plan a whole strategy. Each technique covers another's weakness.

"What does the A* heuristic do, and why is it admissible?"

It estimates remaining cost as laps remaining × fastest possible lap. That can never overestimate the true remaining time, so A* stays optimal while expanding 37% fewer nodes than UCS.

Traps and quick definitions

Things that could catch you out, including inconsistencies in the repo. Know them before someone else points them out.

If this comes up	Say or do this
"How many tests pass?"	Quote the evaluation run: **301 passed, 0 failed, 64.5% coverage** (1 Oct 2026). The README badge says 284 and its testing section says 261; these are out of date.
Unreachable rules	**10 of 32** is current (`domain_validation.json`). Older notes say 14; 4 were fixed.
Calling the test set proof	Don't. Say "one pit lap, so illustrative". Use CV and pooled results for pit claims.
Claiming the DNN wins overall	It wins lap time and loses the pit decision.
The report's "REASONABLE FIT" for the pit DNN	It's what the automated rule outputs. Add that training ROC-AUC 0.68 shows it leans to underfitting.
CV MAE 0.95 for the DNN	From one seed, with early stopping on the scored fold. The 3-seed figure is 1.05 s†.
Replay confusion counts	Quote **precision 0.125 and recall 0.872** only. The four saved counts sum to 940, more than the 901 labelled laps, and we haven't resolved why.
ML pit threshold in the live pipeline	The tuned value is 0.48, but `_run_ml` uses `pipeline.predict`, i.e. 0.5. It's a known gap.
`track_status` scaling	A real defect we found while preparing, not yet fixed in the code. It's better to raise it yourself as a finding (Q9).
Fairness result	Identity features carry little attribution, which isn't a full fairness audit. Per-driver samples are about 11 laps, so they're descriptive only.
Quantum	It's a noiseless simulation. Never claim an advantage.
†-marked numbers	They came from throwaway analysis scripts for this prep, not the committed reports. If asked, say so. Random seeds and small test sets mean they could shift slightly on a rerun.

Definitions in one line

Term	Meaning
MAE / RMSE	Average error in seconds / the same, but penalising big misses more.
R^2	Share of variance explained. 0 equals predicting the mean; negative is worse than that.
PR-AUC	Precision averaged across recall levels. Its baseline is the positive rate, not 0.5.
ROC-AUC	Chance a random positive outranks a random negative.
KS distance	Largest gap between two cumulative distributions, from 0 (same) to 1 (separate).
VIF	How well a feature is predicted by the others. Above 10 signals strong collinearity.
SHAP / LIME	Attribute a prediction to features using game theory / a simple local surrogate model.
Dropout / L2	Randomly disable units in training / penalise large weights. Both reduce overfitting.
Early stopping	Stop when validation loss stops improving and restore the best weights (patience 25 here).
Expanding window	Each fold trains on all earlier laps and validates on the next block.
VSC	Virtual Safety Car: all drivers slow to a set pace, so lap times jump.
Admissible heuristic	Never overestimates remaining cost, which keeps A* optimal.